

ON THE SYMMETRIC MULTIVARIATE NORMAL DISTRIBUTION AND THE ASYMPTOTIC EXPANSION OF A WISHART MATRIX

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SUMMARY

The characteristic function and probability density function of the symmetric multivariate normal distribution which appears as the limiting distribution of a Wishart variate (Anderson (1958)), is derived. Some properties and moments of this symmetric matrix are discussed and an asymptotic approximation to the distribution of

$u = \sqrt{m} \left(\frac{A}{m} - \Sigma \right)$ is derived, where $A \sim W(\Sigma, n)$, $m = n - 2\eta$.

1. Introduction

Anderson (1958) proved that if $A : p \times p \sim W(\Sigma, n)$ then in the limit as $n \rightarrow \infty$, the symmetric matrix

$$(1.1) \quad V : p \times p = \frac{1}{\sqrt{n}} (A - n\Sigma)$$

has a multivariate normal distribution with mean $O : p \times p$ and covariance matrix $\Psi : \frac{1}{2} p(p+1) \times \frac{1}{2} p(p+1)$ with typical element:

$$(1.2) \quad \Psi_{ij,kl} = \text{cov}(V_{ij,kl}) \\ = \sigma_{ik}\sigma_{jl} + \sigma_{il}\sigma_{jk}$$

In Section 3 of this paper we consider this symmetric multivariate normal distribution, with respect to its moments, characteristic function and probability density function.

In Section 4 we develop an asymptotic expansion for $S = \frac{1}{m} A$, where $A \sim W(\Sigma, n)$, $m = n - 2\eta$, in terms of symmetric multivariate normal matrices.

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2. Definitions and identities

Definition 2.1 (Tracy & Dwyer (1969))

If $X : p \times n$, let $\mathbf{x}_c : np \times 1$ denote the column vector formed when the columns of X are stacked in order of appearance into a single column vector.

Let $\mathbf{x}_r : np \times 1$ denote the column vector when the same is done with the row vectors of X .

Notice that if $X : p \times p$ is symmetric then $\mathbf{x}_c = \mathbf{x}_r : p^2 \times 1$.

Defenition 2.2 (Tracy & Singh (1972), Browne (1974))

If $X : p \times p$ is symmetric, let $\mathbf{x}_c^* : \frac{1}{2}p(p+1) \times 1$ denote the column vector formed when the column vectors of X up to and including the element on the main diagonal are stacked into a single column vector, e.g.

$$\mathbf{x}_c^* = [x_{11}, x_{12}, x_{22}, x_{13}, x_{23}, x_{33}, \dots, x_{pp}]$$

Notice that: $\mathbf{x}_c^* = \mathbf{x}_r^*$

Identities

If $X, Y : p \times n$, $A : n \times n$ and $B : p \times p$, the following identities are readily confirmed.

If $Y = BXA$ then:

$$\mathbf{y}_c = (A' \otimes B) \mathbf{x}_c$$

$$\mathbf{y}_r = (B \otimes A') \mathbf{x}_r \quad (\text{Tracy \& Dwyer (1969)})$$

$$(2.1) \quad \text{tr}(XAY'B') = \mathbf{x}_c'(A \otimes B) \mathbf{y}_c$$

$$\text{tr}(X'BYA') = \mathbf{x}_r'(B \otimes A) \mathbf{y}_r$$

$$(2.2) \quad \text{tr}(X'Y) = \mathbf{x}_c' \mathbf{y}_c = \mathbf{y}_c' \mathbf{x}_c$$

$$= \mathbf{x}_r' \mathbf{y}_r = \mathbf{y}_r' \mathbf{x}_r$$

If $X, Y : p \times p$ are symmetric and $A, B : p \times p$ then (2.1) and (2.2) reduce respectively to:

$$(2.3) \quad \text{tr}(XAYB') = \mathbf{x}_c'(A \otimes B) \mathbf{y}_c = \mathbf{x}_r'(A \otimes B) \mathbf{y}_r.$$

$$(2.4) \quad \text{tr}(XY) = \mathbf{x}'_c \mathbf{y}_c = \mathbf{y}'_c \mathbf{x}_c = \mathbf{x}'_r \mathbf{y}_r = \mathbf{y}'_r \mathbf{x}_r.$$

Definition 2.3 (Browne (1974))

Let $\mathbf{K}_p : p^2 \times \frac{1}{2}p(p+1)$ be a matrix with a typical element:

$$(\mathbf{K}_p)_{ij,gh} = \frac{1}{2}(\delta_{ig}\delta_{jh} + \delta_{ih}\delta_{jg}),$$

where $i \leq p, j \leq p, g \leq h \leq p$ and where δ_{ij} denotes Kronecker's delta.

Now introduce the following matrices:

$$(2.5) \quad \mathbf{K}_p^- : \frac{1}{2}p(p+1) \times p^2 = (\mathbf{K}'_p \mathbf{K}_p)^{-1} \mathbf{K}'_p \text{ such that } \mathbf{K}_p^- \mathbf{K}_p = \mathbf{I}_{\frac{1}{2}p(p+1)}$$

and

$$\mathbf{M}_p : p^2 \times p^2 = \mathbf{K}_p \mathbf{K}_p^- \text{ such that } \mathbf{M}_p^2 = \mathbf{M}_p = \mathbf{M}'_p.$$

The following identities, due to Browne are of interest.

$$(2.6) \quad \mathbf{x}_c^* = \mathbf{K}'_p \mathbf{x}_c$$

$$(2.7) \quad \mathbf{x}_c = \mathbf{K}_p^- \mathbf{x}_c^*$$

$$(2.8) \quad \mathbf{M}_p \mathbf{x}_c = \mathbf{x}_c, \mathbf{M}_p \mathbf{K}_p = \mathbf{K}_p \text{ and } \mathbf{K}_p^- \mathbf{M}_p = \mathbf{K}_p^-$$

$$(2.9) \quad \mathbf{M}_p(\mathbf{A} \otimes \mathbf{A}) = (\mathbf{A} \otimes \mathbf{A}) \mathbf{M}_p \text{ for } \mathbf{A} : p \times p$$

Theorem 2.1

If $\mathbf{A} : p \times p$ is non-singular:

$$(2.10) \quad (\mathbf{K}'_p(\mathbf{A} \otimes \mathbf{A})\mathbf{K}_p)^{-1} = \mathbf{K}_p^-(\mathbf{A}^{-1} \otimes \mathbf{A}^{-1})\mathbf{K}_p^{-'} \text{ (Browne (1974))}$$

$$(2.11) \quad (\mathbf{K}_p^-(\mathbf{A} \otimes \mathbf{A})\mathbf{K}_p)^{-1} = \mathbf{K}_p^-(\mathbf{A}^{-1} \otimes \mathbf{A}^{-1})\mathbf{K}_p.$$

Proof

The validity of (2.10) follows by premultiplying both sides by $\mathbf{K}'_p(\mathbf{A} \otimes \mathbf{A})\mathbf{K}_p$.

For the proof of (2.11) we have:

$$(\mathbf{K}_p^-(\mathbf{A} \otimes \mathbf{A})\mathbf{K}_p)^{-1} = ((\mathbf{K}'_p \mathbf{K}_p)^{-1} \mathbf{K}'_p(\mathbf{A} \otimes \mathbf{A})\mathbf{K}_p)^{-1}$$

$$= K_p^-(A^{-1} \otimes A^{-1}) K_p'^{-1} K_p' K_p, \text{ from (2.10)}$$

$$= K_p^-(A^{-1} \otimes A^{-1}) K_p, \text{ from (2.5).}$$

Notation

Let $\lambda_1, \dots, \lambda_p$ be the characteristic roots of a symmetric matrix $A : p \times p$ and $\delta_1, \dots, \delta_p$ be the characteristic roots of a symmetric matrix $B : p \times p$. Let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ and $\Delta = \text{diag}(\delta_1, \dots, \delta_p)$.

Theorem 2.2

If $AB = BA$ with A and B as described above:

$$(2.12) \quad |K_p^-(A \otimes B) K_p| = |K_p^-(\Lambda \otimes \Delta) K_p|.$$

Proof

Since $AB = BA$ there exists an orthogonal matrix $X : p \times p$ such that $X'AX = \Lambda$ and $X'BX = \Delta$. (Bellmann (1970), p. 56.)

Now

$$\begin{aligned} |K_p^-(A \otimes B) K_p| &= |K_p^-(X \otimes X)(\Lambda \otimes \Delta)(X' \otimes X') K_p| \\ &= |K_p^-(X \otimes X) M_p(\Lambda \otimes \Delta) M_p(X' \otimes X') K_p| \text{ from (2.8) and (2.9)} \\ &= |K_p^-(X \otimes X) K_p| |K_p^-(\Lambda \otimes \Delta) K_p| |K_p^-(X' \otimes X') K_p|, \text{ from (2.5)} \\ &= |K_p^-(\Lambda \otimes \Delta) K_p| \text{ from (2.11)} \end{aligned}$$

Theorem 2.3

If $AB = BA$ with A and B as described above:

$$(2.13) \quad |K_p^-(A \otimes B) K_p| = 2^{-\frac{1}{2}p(p+1)} \prod_{i \leq j} (\lambda_i \delta_j + \lambda_j \delta_i).$$

Proof

$$(2.14) \quad |K_p^-(A \otimes B) K_p| = |K_p' K_p|^{-1} |K_p'(\Lambda \otimes \Delta) K_p| \text{ from (2.12) and (2.5)}$$

Since

$$\Lambda \otimes \Delta = \text{diag}(\lambda_1 \delta_1, \lambda_1 \delta_2, \dots, \lambda_1 \delta_p, \lambda_2 \delta_1, \dots, \lambda_2 \delta_p, \dots, \lambda_p \delta_p),$$

$$K'_p(\Lambda \otimes \Delta)K_p = \text{diag}(\lambda_1 \delta_1, \frac{1}{4}(\lambda_1 \delta_2 + \lambda_2 \delta_1), \lambda_2 \delta_2, \dots, \lambda_p \delta_p),$$

with $\lambda_i \delta_i$ in the ii -positions, $i = 1, \dots, p$ and $\frac{1}{4}(\lambda_i \delta_j + \lambda_j \delta_i)$ in the ij, ji -positions, $i < j, i, j = 1, \dots, p$.

Thus

$$(2.15) \quad |K'_p(\Lambda \otimes \Delta)K_p| = 2^{-\frac{1}{2}p(p-1)} 2^{-\frac{1}{2}p(p+1)} \prod_{i \leq j} (\lambda_i \delta_j + \lambda_j \delta_i).$$

If $\Lambda = \Delta = I_p$ in (2.15) then:

$$(2.16) \quad |K'_p K_p| = 2^{-\frac{1}{2}p(p-1)}$$

Substituting (2.15) and (2.16) in (2.14) yields the theorem.

Corollary 2.1

The following results follow from Theorem 2.3

$$(2.17) \quad |K_p^-(A \otimes A)K_p| = \prod_{i \leq j} \lambda_i \lambda_j = |\Lambda|^{p+1} = |A|^{p+1}.$$

$$(2.18) \quad |K'_p(A \otimes 2A)K_p| = 2^p |A|^{p+1}$$

$$(2.19) \quad |K_p^-(A \otimes I_p)K_p| = 2^{-\frac{1}{2}p(p+1)} \prod_{i \leq j} (\lambda_i + \lambda_j).$$

$$|K_p^-(A \otimes I_p + I_p \otimes A)K_p| = \prod_{i \leq j} (\lambda_i + \lambda_j),$$

since $K'_p(A \otimes B)K_p = K'_p(B \otimes A)K_p$, which is a direct consequence of (3.3).

3. The Symmetric Multivariate Normal Distribution

3.1 The characteristic function

If $Z : p \times n \sim N(M, \Sigma \otimes R)$ with $\Sigma, R > 0$, the probability density function (pdf) of Z is given by

$$f(Z) = \frac{\text{etr}(-\frac{1}{2}\Sigma^{-1}(Z-M)R^{-1}(Z-M)')}{(2\pi)^{\frac{1}{2}np} |\Sigma|^{\frac{1}{2}n} |R|^{\frac{1}{2}p}}, \quad -\infty < x < \infty,$$

and the characteristic function is given by

$$(3.1) \quad \phi_Z(T) = E \text{etr}(i T'Z) = \text{etr}(i T'M - \frac{1}{2} T' \Sigma T R), \quad T : p \times n,$$

which can be written in the form:

$$\phi_{\mathbf{z}_r}(\mathbf{t}_r) = E \exp(i \mathbf{t}_r' \mathbf{m}_r - \frac{1}{2} \mathbf{t}_r' (\Sigma \otimes R) \mathbf{t}_r)$$

by using (2.1) and (2.2).

Consider a matrix $Y : p \times p$ such that $\mathbf{y}_c \sim N(\mathbf{n}_c, \Phi)$ where $\Phi : p^2 \times p^2$. Then the vector $\mathbf{x}_c = M_p \mathbf{y}_c \sim N(M_p \mathbf{n}_c, M_p \Phi M_p')$. If $\mathbf{x}_c$ is now written back to the form $X : p \times p$, we notice that $X = X'$ and has a distribution which we call the symmetric normal distribution.

The characteristic function of $\mathbf{y}_c$ is: $\phi_{\mathbf{y}_c}(\mathbf{t}_c) = \exp(i \mathbf{t}_c' \mathbf{n}_c - \frac{1}{2} \mathbf{t}_c' \Phi \mathbf{t}_c)$ and the characteristic function of $\mathbf{x}_c$ is accordingly given by:

$$\phi_{\mathbf{x}_c}(\mathbf{t}_c) = E(\exp i \mathbf{t}_c' M_p \mathbf{y}_c), \text{ where } T : p \times p \text{ is now symmetric.}$$

$$= \exp(i \mathbf{t}_c' M_p \mathbf{n}_c - \frac{1}{2} \mathbf{t}_c' M_p \Phi M_p' \mathbf{t}_c)$$

$$= \exp(i \mathbf{t}_c' \mathbf{m}_c - \frac{1}{2} \mathbf{t}_c' \Phi \mathbf{t}_c), \text{ from (2.8) and}$$

where $\mathbf{m}_c = M_p \mathbf{n}_c$ implies that $M = M'$

If Φ has the covariance structure $\Phi = \Sigma \otimes R$, then:

$$\phi_{\mathbf{x}_c}(\mathbf{t}_c) = \exp(i \mathbf{t}_c' \mathbf{m}_c - \frac{1}{2} \mathbf{t}_c' \Sigma \otimes R \mathbf{t}_c)$$

i.e.

$$(3.2) \quad \phi_X(T) = \text{etr}(i T M - \frac{1}{2} T \Sigma T R)$$

where T, Σ and $R : p \times p$ are symmetric and $\Sigma R = R \Sigma$.

Equation (3.1) implies that the covariance matrix of $Z : p \times n$ is $\Sigma \otimes R$. If $X : p \times p$ is symmetric, then X contains only $\frac{1}{2} p(p+1)$ different elements, thus (3.2) does not imply that $\Sigma \otimes R$ is the covariance matrix of X . To obtain the covariance matrix we will write (3.2) in a form where only the different elements of X appear:

$$\phi_X(T) = \phi_{\mathbf{x}_c}(T) = E \exp(i \mathbf{t}_c' \mathbf{x}_c)$$

$$= E \exp(i \mathbf{t}_c' K_p^{-1} K_p' \mathbf{x}_c^*), \text{ from (2.6)}$$

$$= E \exp (i \tau_c^* x_c^*)$$

$$= \phi_{x_c^*}(\tau_c^*)$$

$$\text{where } \tau_c^{*'} = ((K_p' K_p)^{-1} \mathbf{t}_c^*)' = [t_{11}, 2t_{12}, t_{22}, 2t_{13}, 2t_{23}, t_{33}, \dots, t_{pp}]$$

If the same is done with the right-hand side of (3.2) we obtain

$$\phi_X(T) = \phi_{x_c}(\mathbf{t}_c) = \exp (i \mathbf{t}_c' \mathbf{m}_c - \frac{1}{2} \mathbf{t}_c' (\Sigma \otimes R) \mathbf{t}_c)$$

i.e.

$$\phi_{x_c^*}(\tau_c^*) = \exp (i \tau_c^{*'} \mathbf{m}_c^* - \frac{1}{2} \tau_c^{*'} K_p' (\Sigma \otimes R) K_p \tau_c^*)$$

This is the characteristic function of a $\frac{1}{2} p(p+1)$ component vector x_c^* with a multivariate normal distribution:

$$x_c^* \sim N(m_c^*, K_p' (\Sigma \otimes R) K_p)$$

Thus we will denote the distribution of X as:

$$X = X' \sim N(M, K_p' (\Sigma \otimes R) K_p)$$

A typical element of the covariance matrix

$$K_p' (\Sigma \otimes R) K_p : \frac{1}{2} p (p+1) \times \frac{1}{2} p (p+1) \text{ is given by}$$

$$(3.3) \text{Cov}(X_{ij}, X_{kl}) = \frac{1}{4} (\sigma_{ik} r_{jl} + \sigma_{jk} r_{il} + \sigma_{il} r_{jk} + \sigma_{jl} r_{ik})$$

Examples

If $\phi_X(T) = \text{etr}(-T\Sigma T^2)$ then:

$$X = X' \sim N(O, K_p' (2\Sigma \otimes \Sigma) K_p)$$

$$\text{and } \text{Cov}(X_{ij}, X_{kl}) = \sigma_{ik} \sigma_{jl} + \sigma_{il} \sigma_{jk}$$

which agrees with (1.2) as derived by Anderson (1958), p. 75.

$$\text{If } \phi_X(T) = \text{etr}(-T^2) \text{ then}$$

$$X = X' \sim N(O, K_p' (2I_p \otimes I_p) K_p)$$

$$\text{and } \text{Cov}(X_{ij}, X_{kl}) = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$$

$$\text{i.e. } \text{Cov}(X_{ij}, X_{kl}) = 0 \text{ if } i, j \neq k, l \text{ or } i, j \neq 1, k$$

$$\text{and } \text{Var}(X_{ii}) = 2, \text{Var}(X_{ij}) = 1$$

3.2 The probability density function

$$\text{Let } X = X' \sim N(M, K_p' (\Sigma \otimes R) K_p) \text{ i.e. } \mathbf{x}_c^* \sim N(\mathbf{m}_c^*, K_p' (\Sigma \otimes R) K_p)$$

then the probability density function (pdf) can be written as:

$$f(\mathbf{x}_c^*) = \frac{\exp(-\frac{1}{2}(\mathbf{x}_c^* - \mathbf{m}_c^*)'(K_p'(\Sigma \otimes R)K_p)^{-1}(\mathbf{x}_c^* - \mathbf{m}_c^*))}{(2\pi)^{\frac{1}{4}p(p+1)} |K_p'(\Sigma \otimes R)K_p|^{\frac{1}{2}}}$$

$$\text{where } -\infty < \mathbf{x}_c^* < \infty$$

If $R = k\Sigma$, where $k \neq 0$ is a scalar constant, the pdf can be simplified as:

$$f(\mathbf{x}_c) = \frac{\exp(-\frac{1}{2}(\mathbf{x}_c - \mathbf{m}_c)'(\Sigma^{-1} \otimes k^{-1}\Sigma^{-1})(\mathbf{x}_c - \mathbf{m}_c))}{(2\pi)^{\frac{1}{4}p(p+1)} |K_p'(\Sigma \otimes k\Sigma)K_p|^{\frac{1}{2}}}$$

$$\text{where } -\infty < \mathbf{x}_c < \infty$$

Thus

$$f(X) = \frac{\text{etr}(-\frac{1}{2k}\Sigma^{-1}(X-M)\Sigma^{-1}(X-M))}{(2\pi)^{\frac{1}{4}p(p+1)} |K_p'(\Sigma \otimes k\Sigma)K_p|^{\frac{1}{2}}}$$

$$\text{where } -\infty < X < \infty.$$

Special cases

$$\text{If } X = X' \sim N(M, K_p'(2\Sigma \otimes \Sigma)K_p):$$

$$f(X) = \frac{\text{etr}(-\frac{1}{4}(\Sigma^{-1}(X-M))^2)}{(2\pi)^{\frac{1}{4}p(p+1)} 2^{\frac{1}{2}p} |\Sigma|^{\frac{1}{2}(p+1)}}, -\infty < X < \infty, \text{ (from 2.18)}$$

$$\text{If } X = X' \sim N(M, K_p'(2I_p \otimes I_p)K_p):$$

$$f(X) = \frac{\text{etr}(-\frac{1}{4}(X-M)^2)}{(2\pi)^{\frac{1}{4}p(p+1)} 2^{\frac{1}{2}p}}, \quad -\infty < X < \infty$$

If $X = X' \sim N(M, K_p'(I_p \otimes I_p) K_p)$ then:

$$f(X) = \frac{\text{etr}(-\frac{1}{2}(X-M)^2)}{\pi^{\frac{1}{4}p(p+1)} 2^{\frac{1}{2}p}}, \quad -\infty < X < \infty$$

the following theorem is of interest.

Theorem 3.1

If $X = X' \sim N(M, K_p'(\Sigma \otimes R) K_p)$, $A : p \times p$ is a symmetric matrix such that $\Sigma AR = RA\Sigma$ and $h(X)$ defines an elementary symmetric function of X , then:

$$\begin{aligned} E \text{etr}(AX) h(X) \\ = \text{etr}(AM + \frac{1}{2}\Sigma ARA) E(h(\ddot{X})) \end{aligned}$$

where

$$\ddot{X} = \ddot{X}' \sim N(M + \Sigma AR, K_p'(\Sigma \otimes R) K_p)$$

Proof:

$$\begin{aligned} E(h(x) \text{etr}(AX)) &= E(h(x_c) \exp(x_c' a_c)) \\ &= E(h(x_c^*) \exp(x_c^* K_p^- K_p^{-'} a_c^*)) \\ &= K \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left\{ -\frac{1}{2} [(x_c^* - m_c^*)'(K_p'(\Sigma \otimes R) K_p)^{-1} (x_c^* - m_c^*) \right. \\ &\quad \left. - 2a_c^{*'} K_p^- K_p^{-'} x_c^*] \right\} h(x_c^*) dx_c^*, \end{aligned}$$

$$\text{where } K^{-1} = (2\pi)^{\frac{1}{4}p(p+1)} |K_p'(\Sigma \otimes R) K_p|^{-\frac{1}{2}}$$

$$= K \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left\{ -\frac{1}{2} ((x_c^* - m_c^* - K_p'(\Sigma \otimes R) K_p K_p^- K_p^{-'} a_c^*)(K_p'(\Sigma \otimes R) K_p)^{-1} \right.$$

$$\begin{aligned}
& \cdot (\mathbf{x}_c^* - \mathbf{m}_c^* - \mathbf{K}_p'(\Sigma \otimes \mathbf{R}) \mathbf{K}_p \mathbf{K}_p^{-1} \mathbf{a}_c^*) \} h(\mathbf{x}_c^*) d\mathbf{x}_c^* \\
& \cdot \exp \{ \mathbf{a}_c^{*'} \mathbf{K}_p^{-1} \mathbf{K}_p^{-1'} \mathbf{m}_c^* + \frac{1}{2} \mathbf{a}_c^{*'} \mathbf{K}_p^{-1} \mathbf{K}_p^{-1'} \mathbf{K}_p'(\Sigma \otimes \mathbf{R}) \mathbf{K}_p \mathbf{K}_p^{-1} \mathbf{a}_c^* \} \\
& = \exp \{ \mathbf{a}_c' \mathbf{m}_c + \frac{1}{2} \mathbf{a}_c'(\Sigma \otimes \mathbf{R}) \mathbf{a}_c \} E(h(\mathbf{x}_c^*)),
\end{aligned}$$

where $\mathbf{x}_c^* \sim N(\mathbf{m}_c^* + \mathbf{z}_c^*, \mathbf{K}_p'(\Sigma \otimes \mathbf{R}) \mathbf{K}_p)$,

$$\begin{aligned}
\mathbf{z}_c^* &= \mathbf{K}_p' \mathbf{z}_c \text{ and } \mathbf{z}_c = (\Sigma \otimes \mathbf{R}) \mathbf{a}_c \quad (\text{see (2.8)}) \\
&= \text{etr} \left(\mathbf{A}\mathbf{M} + \frac{1}{2} \Sigma \mathbf{A}\mathbf{R}\mathbf{A} \right) E(h(\tilde{\mathbf{x}})),
\end{aligned}$$

where $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}' \sim N(\mathbf{M} + \Sigma \mathbf{A}\mathbf{R}, \mathbf{K}_p'(\Sigma \otimes \mathbf{R}) \mathbf{K}_p)$

Corollary: If $\mathbf{A} = \mathbf{i}\mathbf{i}'$ and $h(\mathbf{X}) = 1$, Theorem 3.1 serves as a proof that (3.2) is indeed the characteristic function of $\mathbf{X} = \mathbf{X}' \sim N(\mathbf{M}, \mathbf{K}_p'(\Sigma \otimes \mathbf{R}) \mathbf{K}_p)$.

3.3 On expected values of $\mathbf{X}$, where $\mathbf{X} = \mathbf{X}' \sim N(\mathbf{M}, \mathbf{K}_p'(\Sigma \otimes \mathbf{R}) \mathbf{K}_p)$

Using (3.3) and the fact that:

$$\begin{aligned}
E(X_{ij} X_{kl}) &= \text{Cov}(X_{ij}, X_{kl}) + m_{ij} m_{kl}, \\
E(X_{ij} X_{kl} X_{mn}) &= m_{ij} \text{Cov}(X_{kl}, X_{mn}) + m_{kl} \text{Cov}(X_{ij}, X_{mn}) \\
&\quad + m_{mn} \text{Cov}(X_{ij}, X_{kl}) + m_{ij} m_{kl} m_{mn}
\end{aligned}$$

and

$$\begin{aligned}
E(X_{ij} X_{kl} X_{mn} X_{pq}) &= \text{Cov}(X_{ij}, X_{pq}) \text{Cov}(X_{kl}, X_{mn}) \\
&\quad + \text{Cov}(X_{ij}, X_{kl}) \text{Cov}(X_{pq}, X_{mn}) + \text{Cov}(X_{ij}, X_{mn}) \text{Cov}(X_{pq}, X_{kl}) \\
&\quad + m_{ij} m_{kl} \text{Cov}(X_{pq}, X_{mn}) + m_{ij} m_{mn} \text{Cov}(X_{pq}, X_{kl}) \\
&\quad + m_{kl} m_{mn} \text{Cov}(X_{ij}, X_{pq}) + m_{ij} m_{pq} \text{Cov}(X_{kl}, X_{mn}) \\
&\quad + m_{pq} m_{kl} \text{Cov}(X_{ij}, X_{mn}) + m_{pq} m_{mn} \text{Cov}(X_{ij}, X_{kl}) + m_{ij} m_{kl} m_{mn} m_{pq},
\end{aligned}$$

we obtain for B, C and $D : p \times p$ constant and symmetric.

$$(3.4) \quad E(XBX) = \frac{1}{4} (\Sigma BR + RB\Sigma + \Sigma(\text{tr} BR) + R(\text{tr} B\Sigma)) + MBM,$$

$$(3.5) \quad E(XBXCX) = \frac{1}{4} \{MB\Sigma CR + MBRC\Sigma + \Sigma CMBR + RCMB\Sigma$$

$$+ \Sigma BRCM + RB\Sigma CM + MBR(\text{tr} C\Sigma)$$

$$+ MB\Sigma(\text{tr} CR) + R(\text{tr} CMB\Sigma) + \Sigma(\text{tr} CMBR)$$

$$+ RCM(\text{tr} B\Sigma) + \Sigma CM(\text{tr} BR)\} + MBMCM,$$

$$(3.6) \quad E(XBXC DX) = \frac{1}{16} \{ \Sigma DRC\Sigma BR + RDRC\Sigma B\Sigma + \Sigma D\Sigma CRBR$$

$$+ RD\Sigma CRB\Sigma + \Sigma BRCS\Delta R + \Sigma BRCD\Sigma + RB\Sigma C\Delta R$$

$$+ RB\Sigma CRD\Sigma + \Sigma CRDRB\Sigma + \Sigma C\Sigma DRBR + RD\Sigma B\Sigma CR$$

$$+ RDRC\Sigma B\Sigma + RD\Sigma C\Sigma BR + RDRB\Sigma C\Sigma + \Sigma D\Sigma BRCR$$

$$+ \Sigma DRCRB\Sigma + \Sigma D\Sigma CRBR + \Sigma DRBRC\Sigma + RCRD\Sigma B\Sigma$$

$$+ RC\Sigma D\Sigma BR$$

$$+ R(\text{tr} \Sigma DRC\Sigma B + \text{tr} \Sigma D\Sigma CRB + (\text{tr} \Sigma DRB)(\text{tr} C\Sigma))$$

$$+ (\text{tr} \Sigma D\Sigma B)(\text{tr} CR))$$

$$+ \Sigma(\text{tr} (RDRC\Sigma B + \text{tr} RD\Sigma CRB + (\text{tr} RDRB)(\text{tr} C\Sigma))$$

$$+ (\text{tr} RD\Sigma B)(\text{tr} CR))$$

$$+ RC\Sigma(\text{tr} BRD\Sigma + (\text{tr} B\Sigma)(\text{tr} DR)) + \Sigma C\Sigma(\text{tr} RDRB$$

$$+ (\text{tr} BR)(\text{tr} DR))$$

$$+ \Sigma CR(\text{tr} BRD + (\text{tr} BR)(\text{tr} D\Sigma)) + RCR(\text{tr} B\Sigma D\Sigma$$

$$\begin{aligned}
& + (\text{tr } B\Sigma)(\text{tr } D\Sigma)) \\
& + (\Sigma DRBR + RDRB\Sigma) \text{tr } C\Sigma + (\Sigma D\Sigma BR + RD\Sigma B\Sigma) \text{tr } CR \\
& + (RC\Sigma DR + RCRD\Sigma) \text{tr } B\Sigma + (\Sigma C\Sigma DR + \Sigma CRD\Sigma) \text{tr } BR \\
& + (\Sigma BR CR + RB\Sigma CR)\text{tr } D\Sigma + (\Sigma BR C\Sigma + RB\Sigma C\Sigma) \text{tr } DR \} \\
& + \frac{1}{4} \{ MB\Sigma CRDM + MBRC\Sigma DM + \Sigma BR CDM + RB\Sigma CDM \\
& + \Sigma CMBRDM + RCMB\Sigma DM + MBMC\Sigma DR + MBMCRD\Sigma \\
& + \Sigma MB\Sigma DMCR + MBRDMC\Sigma + \Sigma DMCMBR + RDMCM B\Sigma \\
& + MBRDM(\text{tr } C\Sigma) + MB\Sigma DM(\text{tr } CR) + RCMDM(\text{tr } B\Sigma) \\
& + \Sigma CMDM(\text{tr } BR) \\
& + MBMC\Sigma(\text{tr } DR) + MBMCR(\text{tr } D\Sigma) + RDM(\text{tr } BMC\Sigma) \\
& + DM(\text{tr } MCRB) \\
& + MB\Sigma(\text{tr } RCMD) + MBR(\text{tr } \Sigma CMD) + \Sigma(\text{tr } RDMCM B) \\
& + R(\text{tr } \Sigma DMCM B) \} \\
& + MBMCMDM.
\end{aligned}$$

In particular

If $B = \Sigma^{-1}$, $R = 2\Sigma$, $M = 2i\Sigma T$ and if we premultiply by iT , (3.4) becomes:

$$(3.7) \quad E(iTX\Sigma^{-1}X) = (p+1) iT\Sigma + 4(it\Sigma)^3.$$

4. An Asymptotic Expansion for Wishart Matrices

To improve Anderson's (1958) limiting distribution for $A \sim W(\Sigma, n)$ (see (1.1)) we approximate the distribution of $\frac{A}{m}$ by the distribution of

$$A_m(V) = \Sigma + \frac{1}{\sqrt{m}} V^{(1)} + \frac{1}{m} V^{(2)} + \frac{1}{m^{3/2}} V^{(3)},$$

in the sense that their asymptotically expanded characteristic functions will be the same up to the order terms considered.

Let $V^{(1)} = V = V' \sim N(O, K_p'(\Sigma \otimes 2\Sigma) K_p)$ and suppose $V^{(2)}$ and $V^{(3)}$ are matrix functions of V and $m = n - 2\eta$, where η is an arbitrary constant.

Equivalently the distribution of:

$$U = \sqrt{m} \left(\frac{A}{m} - \Sigma \right),$$

is approximated by the distribution of:

$$B_m(V) = V^{(1)} + \frac{1}{\sqrt{m}} V^{(2)} + \frac{1}{m} V^{(3)}.$$

The characteristic function of U is given by Sugiura (1973):

$$\begin{aligned} \phi_U(T) &= E \operatorname{etr}(iT U) \\ &= \operatorname{etr}(-(T\Sigma)^2) \left\{ 1 + \frac{1}{\sqrt{m}} (2\eta \operatorname{tr}(iT\Sigma) + \frac{4}{3} \operatorname{tr}(iT\Sigma)^3) \right. \\ &\quad + \frac{1}{m} (2\eta \operatorname{tr}(iT\Sigma)^2 + 2\eta^2 \operatorname{tr}^2(iT\Sigma) + 2\operatorname{tr}(iT\Sigma)^4) \\ &\quad \left. + \frac{8}{3} \eta \operatorname{tr}(iT\Sigma) \operatorname{tr}(iT\Sigma)^3 + \frac{8}{9} \operatorname{tr}^2(iT\Sigma)^3 \right\} + O(m^{-3/2}). \end{aligned}$$

Suppose $V^{(2)}$ and $V^{(3)}$ are matrix functions of V , denoted as

$$V^{(2)} = f(V) \text{ and } V^{(3)} = g(V).$$

Using a Taylor expansion, the characteristic function of $B_m(V)$ is:

$$\begin{aligned} \phi_{B_m(V)}(T) &= E \operatorname{etr}(iT V) \left\{ 1 + \frac{1}{\sqrt{m}} \operatorname{tr}(iT f(V)) \right. \\ &\quad \left. + \frac{1}{m} \left(\frac{1}{2} \operatorname{tr}^2(iT f(V)) + \operatorname{tr}(iT g(V)) \right) + O_p(m^{-3/2}) \right\}. \end{aligned}$$

Comparing the characteristic functions of U and $B_m(V)$, we must have:

$$(4.1) \quad E \operatorname{etr}(iT V) \operatorname{tr}(iT f(V)) = \{2\eta \operatorname{tr}(iT\Sigma) + \frac{4}{3} \operatorname{tr}(iT\Sigma)^3\} \operatorname{etr}(-(T\Sigma)^2),$$

$$\begin{aligned} (4.2) \quad E \operatorname{etr}(iT V) (\operatorname{tr}(iT g(V)) + \frac{1}{2} \operatorname{tr}^2(iT f(V))) &= \{2\eta \operatorname{tr}(iT\Sigma)^2 + 2\eta^2 \operatorname{tr}^2(iT\Sigma) \\ &\quad + 2 \operatorname{tr}(iT\Sigma)^4 + \frac{8}{3} \eta \operatorname{tr}(iT\Sigma) \operatorname{tr}(iT\Sigma)^3 + \frac{8}{9} \operatorname{tr}^2(iT\Sigma)^3\} \operatorname{etr}(-(T\Sigma)^2) \end{aligned}$$

According to Theorem 3.1 and from (3.7) it follows that:

$$\begin{aligned} E \operatorname{etr}(iTV) \operatorname{tr}\left(\frac{1}{3} iTV\Sigma^{-1}V\right) \\ = \operatorname{etr}(-(T\Sigma)^2) E \operatorname{tr}\left(\frac{1}{3} iT\ddot{V}\Sigma^{-1}\ddot{V}\right), \end{aligned}$$

where $\ddot{V} = \ddot{V}' \sim N(2i\Sigma T\Sigma, K_p'(2\Sigma \otimes \Sigma) K_p)$

$$= \operatorname{etr}(-(T\Sigma)^2) \left\{ \frac{(p+1)}{3} \operatorname{tr}(iT\Sigma) + \frac{4}{3} \operatorname{tr}(iT\Sigma)^3 \right\},$$

which yields (4.1) with $\eta = \frac{p+1}{6}$ and $f(V) = \frac{1}{3} V\Sigma^{-1}V$.

Thus the distribution of $\frac{A}{m}$ is approximated by the distribution of:

$$(4.3) \quad A_m(V) = \Sigma + \frac{1}{\sqrt{m}} V + \frac{1}{3m} V\Sigma^{-1}V,$$

where $V = V' \sim N(0, K_p'(2\Sigma \otimes \Sigma) K_p)$

and $m = n - (p+1)/3$.

To find an expression for $g(V)$ which yields (4.2) proved to be a very laborious task and will be given in a further communication.

Special Case

If $p = 1$ and $\sigma^2 = 1$ it is possible to find an asymptotic expansion for a χ_n^2 variable up to terms of order n^{-1} .

Now the distribution of

$$U = \sqrt{m} \left(\frac{\chi_n^2}{m} - 1 \right) \text{ is approximated by the distribution of}$$

$$B_m(V) = V^{(1)} + \frac{1}{\sqrt{m}} V^{(2)} + \frac{1}{m} V^{(3)},$$

where $V^{(1)} = V \sim N(0, 2)$,

$$V^{(2)} = \frac{1}{3} V^2, \text{ according to (4.3)}$$

and $m = n - 2\eta$ with $\eta = \frac{1}{3}$.

Now $V^{(3)} = g(V)$ must be a function of V such that:

$$E(e^{itV})(itV^{(3)} + \frac{(it)^2}{18} V^4)$$

$$\begin{aligned}
&= e^{-t^2} \{2\eta(it)^2 + 2\eta^2(it)^2 + 2(it)^4 + \frac{8}{9}(it)^6 + \frac{8}{3}\eta(it)^4\} \\
&= e^{-t^2} \frac{1}{9} \{8(it)^6 + 26(it)^4 + 8(it)^2\}.
\end{aligned}$$

Using the central moments of V it follows that

$$v^{(3)} = \frac{1}{72} (26V^3 - 24V^2 - 148V + 48)$$

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